Problem Editorial Submissions Comments

Seating Arrangement

Difficulty: **Easy** Accuracy: **49.85%** Submissions: **59K+** Points: **2**

Given an integer **k** representing the number of people to be seated and an array **seats[]**, where **0** denotes an empty seat and **1** denotes an occupied seat.

Determine whether it is possible to seat all **k** people such that no two occupied seats are adjacent (including newly seated people).

Examples:

Input: k = 2, seats[] = [0, 0, 1, 0, 0, 0, 1]
Output: true
Explanation: The two people can sit at index 0 and 4.

Input: k = 1, seats[] = [0, 1, 0]
Output: false
Explanation: There is no way to get a seat for one person.

Input: k = 0, seats[] = [0, 0, 0, 1, 1]
Output: false
Explanation: The seating arrangement already contains two

Python3 Start Timer

```python
class Solution:
    def canSeatAllPeople(self, k, seats):
        n = len(seats)

        # Check if current arrangement is already invalid
        for i in range(n - 1):
            if seats[i] == 1 and seats[i + 1] == 1:
                return False

        count = 0

        for i in range(n):
            if seats[i] == 0:
                left_empty = (i == 0) or (seats[i - 1] == 0)
                right_empty = (i == n - 1) or (seats[i + 1] =

                if left_empty and right_empty:
                    seats[i] = 1
                    count += 1

        return count >= k
```

Custom Input Compile & Run Submit